All 24 D-Flow choices are frozen

The study is unfinished. All eight Task 2 D-Flow conditions have completed literal top-three fresh40 confirmation for each independently trained model. QGF and FK each have three frozen choices. IPG remains unqualified, so the required all 33 barrier is incomplete and no new main heldout-100 test has started.

Own-best condition	lambda	Seed 11	Seed 23	Seed 37
Regular / raw Q	0	35/40	40/40	37/40
Regular / raw Q	1	28/40	40/40	40/40
Stochastic / raw Q	0	40/40	40/40	39/40
Stochastic / raw Q	1	37/40	40/40	40/40
Regular / averaged Q	0	38/40	40/40	37/40
Regular / averaged Q	1	25/40	40/40	40/40
Stochastic / averaged Q	0	40/40	40/40	40/40
Stochastic / averaged Q	1	35/40	40/40	40/40
Original QGF	-	40/40	40/40	40/40
Original FK	-	40/40	40/40	40/40
Original IPG	-	unqualified	unqualified	unqualified

Interpretation: these confirmation results do not show tuned D-Flow outperforming tuned QGF. They are independently selected settings, not matched-parameter causal effects of Q averaging and not heldout performance. The three training seeds are independent models; paired rollout starts are paired observations, not more training seeds.

QGF selected (M,kappa): seed 11 (64,16), seed 23 (8,320), seed 37 (16,16). FK selected kappa80,N32,noise .02 with M64/32/16. Equal numeric kappa across methods does not imply equal accumulated guidance.

Original DF24 aggregate marker:19d8af37; independent terminal audit:39c0d55d. The freeze queried no model, sampler or environment. Its24 choices retain all72 literal alternatives. No hidden fourth candidate was substituted.

Every selected D-Flow tuple

All rows use identity surrogate guidance, full-F8 endpoints and T=1. I is latent updates, K is Q neighbors. The final column retains the original literal three fresh40 outcomes; the selected score can be from a later-ranked candidate.

Seed	Mode/Q	L	kappa	eta	I	sigma	K	CONF	Literal3 /40
11	reg/raw	0	0.1	0.1	256	-	-	35/40	35/31/21
11	reg/raw	1	40	0.03	256	-	-	28/40	24/28/0
11	sto/raw	0	5	0.1	128	-	-	40/40	40/24/22
11	sto/raw	1	5	0.1	128	-	-	37/40	37/23/10
11	reg/avg	0	1	0.1	256	0.003	16	38/40	36/38/21
11	reg/avg	1	64	0.03	256	0.5	4	25/40	24/25/24
11	sto/avg	0	5	0.1	128	0.003	16	40/40	40/29/24
11	sto/avg	1	5	0.03	128	0.003	16	35/40	35/26/26
23	reg/raw	0	5	0.1	256	-	-	40/40	40/40/40
23	reg/raw	1	64	0.03	256	-	-	40/40	40/40/40
23	sto/raw	0	80	0.03	256	-	-	40/40	40/40/22
23	sto/raw	1	80	0.01	128	-	-	40/40	40/33/30
23	reg/avg	0	16	0.1	256	0.3	4	40/40	40/40/40
23	reg/avg	1	80	0.03	128	0.5	16	40/40	40/40/40
23	sto/avg	0	64	0.1	128	0.5	4	40/40	40/40/40
23	sto/avg	1	80	0.1	256	0.01	4	40/40	40/40/40
37	reg/raw	0	0.1	0.1	256	-	-	37/40	37/34/25
37	reg/raw	1	5	0.1	128	-	-	40/40	40/22/0
37	sto/raw	0	5	0.1	128	-	-	39/40	39/39/32
37	sto/raw	1	5	0.1	128	-	-	40/40	40/24/4
37	reg/avg	0	0.1	0.1	256	0.003	4	37/40	37/36/23
37	reg/avg	1	5	0.03	256	0.003	4	40/40	40/26/24
37	sto/avg	0	3	0.1	256	0.003	4	40/40	40/34/30
37	sto/avg	1	5	0.03	128	0.003	4	40/40	40/25/25

Full machine-readable tuples, configuration IDs, model identities, source paths and alternatives are in RESULTS.json and DF24_SELECTED_TUPLES.csv. All 27,720 TRAIN scores are preserved separately; they are not native success episodes.

DF ranges: 20 kappas from 0 to 80; eta .001/.003/.005/.01/.03/.1; I1/2/4/8/16/32/64/128/256; averaged-Q sigma .003/.01/.03/.06/.1/.2/.3/.5 and K4/16. Hierarchical 9,240 settings/model, not 73,440 Cartesian. Historical I512 and FD32 failures remain failures.

Harder Cube: trained, not yet evaluated

The CubeTripleTask3 pilot has 46-dimensional observations and 25-dimensional action chunks: five consecutive controls of five coordinates. It uses unaugmented seed 11 models, 128x128 networks, H5, F8 and T=1. Its local 1M-transition data slice is a limited-data adaptation, not the paper 100M-data replication.

Completed stage	Actual outcome	Outer wall, once
Fresh BC	50,000 BC/Adam updates; final actor parent 91a8	236.6263 s
Goal-specific IQL	50,000 Q/V updates; full actor state and RNG frozen; final 2935	260.3903 s
E1 callbacks	69 checks PASS; actual H5/flat25 model interface	7.2159 s
Controlled H5 comparison	31 checks PASS; 44 snapshots / 40 external model-free controls	120.6904 s
Official-reset R0 observer	19 checks PASS; 4 reset bodies / 8 internal controls / 0 external	9.9605 s
First E2a sampler gate	FAILED in first graph inventory; 0 sampler or FD calls	6.7059 s

No eight-condition Cube policy comparison has run. Training completion and callback/queue checks do not establish native competence or Q accuracy. The final IQL checkpoint restores the full actor, critics, target critics, value function, optimizer state and RNG; no Task1 critic is reused for Task3 reward.

TRAIN reward is per-cube goal count minus 3: 927,047 rows at -3; 71,460 at -2; 1,493 at -1; zero joint-goal rows. There are 996,000 full-H5 eligible tuples, with 4,000 short tails excluded. This is descriptive coverage, not proof of impossibility, competence or true Q error. A useful primitive policy need not achieve the whole goal during a BC readiness check.

The sampler correction remains prospective

The first E2a attempt stopped because the metadata visitor treated ClosedJaxpr as a bare Jaxpr and read an absent constvars field. One trace returned; no dtype inventory completed, no sampler/FD query ran, and reference64 was not launched. Its 49 earlier PASS predicates apply only to earlier gates. A separate strict visitor correction passed both SOURCE reviews; it remains unrun. No tolerance, numerical formula, grid, dtype policy or random bank was relaxed.

Hard cases: failures are preserved

Known-Q fixtures in dimensions 8, 32 and 128

All 596 finite engineering configurations executed: 585 PASS / 11 FAIL. The gate remains unqualified. All 11 are strong-guidance original-QGF formula-parity failures; finite, replay, source and action-bound checks passed. These are synthetic numerical checks, not robotic return measurements or demonstrated IQL Q-error failures.

The large case is dimension 32, legitimate narrow-good Q, M32 / kappa 80: final production/reference action difference 1.86511415, affecting 625/640 coordinates. Its kappa-zero control differs by only 1.19e-7 and passes. A compact capture reproduced the old public outputs before exporting intermediates. Saved byte flags do not measure magnitudes or expose actual consumer memory; a unique compiler/hardware cause or repair is not established.

Original IPG is still unqualified

The independent materialized-reference B144 gate has 134 PASS / 10 FAIL, with 10 particle-close failures and one overlapping action-close failure. Old residual failures (33), missing learned cases (15), dtype/lowering failures and both ridge domains remain preserved. Passing residual/source/noise checks does not waive trajectory checks. There is no C/native IPG release.

Task 5: tuned QGF did not generally collapse

Training seed	Selected kappa	M	Fresh CONF
11	96	16	40/40
23	2	64	36/40
37	80	32	40/40

The selected Task 5 QGF models scored 40/40, 36/40 and 40/40. Four seed-23 failures are real, but do not establish generic QGF collapse or its cause. Expanded Task 5 DF engineering retains four latent-check failures and an inconclusive diagnostic; no expanded native DF comparison is admitted.

The standard full-route withheld-action probe had zero joint withheld-action exposure. A custom right-start probe exposed that region, but used a custom reset and did not establish true Q error or an averaging repair. Poor settings, synthetic parity failures, custom resets and independently tuned native task outcomes are different evidence.

Reset evidence and Q averaging

Equal observations did not prove paired simulator states

R0 used two original registered environment-only builds, including their implicit checker resets, followed by two explicit resets with the same environment seed. The original fresh Box generators were untouched. It recorded four native reset bodies, eight internal controls and zero external controls; 200 physics substeps are source-implied from returned 25-step parameters, not direct C-kernel observations.

Both returned 46D observation descriptors have the same saved payload SHA. Nine broader recognized-state SHA leaves differ: integration state, selected force/acceleration fields, cached goal/control data and the target effector pose. Volatile logging clocks were retained separately and excluded from this comparison. This finite observation does not identify a unique cause, numeric magnitude or future performance effect.

The independent auditor recomputed JSON consistency and the nine descriptor differences, checked opaque archive hashes and source lineage, and verified cleanup. It did not decode snapshot payloads or replay physics. Future Cube comparisons should be labelled unpaired unless a separate protocol establishes the required full-state matching. R0 is not a policy-performance or restoration qualification.

One endpoint, K critic neighbors

$$\bar{Q}(s,a) = \frac{1}{K} \sum_j Q(s, \mathrm{clip}(a + \sigma \epsilon_j))$$

The same flow endpoint can receive K nearby critic queries. K does not mean K flow trajectories and adds no physical transitions. Sigma sets neighborhood width; K sets finite sample count. The perturbation bank stays fixed throughout the action generation, including all I latent updates. K4 and K16 banks need not be nested.

Averaging may soften a narrow false peak if neighboring predictions are reasonable, may leave broad errors unchanged, or may blur a genuinely narrow good action. It cannot guarantee unseen-Q repair. Latent stochastic noise is separate. Lambda 1 is latent prior pull, not an L1 norm/support detector. Identity guidance is a surrogate; learned-V residuals and MC Qpi discrepancies are not ground-truth IQL Q errors.

Partial checkpoint: scope and resources

The requested study is unfinished. All 24 DF choices plus QGF3 and FK3 are fixed, but IPG3 is unqualified. The strict all-33 requirement remains in force. No new main heldout-100 test, matched-kappa curve campaign or Cube eight-condition utility comparison has been released.

Closed quantity	Amount
New native VAL/CONF episodes	30,360
Evaluated policy active action rows	16,185,534
New main heldout episodes	0
Measured scientific outer-parent wall	45295.281965003 s, once
Separate unknown old-admission charge	600 s allowance; not measured actual wall
Active caps / remaining of 86,400 s	0 / 40504.718034997 s

Model-free H5/R0 reset and engineering controls have separate counters; they are not policy VAL/CONF episodes or evaluated policy action rows. Administration, child, watchdog and sampler clocks are not added to the measured scientific outer-parent clock. The initial 24 worker-hours are a bounded tranche, not a whole-study completion promise.

Remaining work and current limits

Remaining scope includes IPG qualification/selections, the all-33 barrier and fresh main 100, 540 matched-kappa points targeting 100 episodes each in declared 20-episode increments, and unfinished Task 5/Cube/higher-dimensional work. No hidden fourth candidate, grid/episode cut, heldout tuning or forced winner is used.

An optional narrower main-test proposal exists for 30 qualified choices plus BC3 while reporting IPG as unqualified. It has not been adopted and creates no test authority. Current guards prohibit expensive starts after 14:20 UTC. Any later calendar requires prospective protocol/source reviews, checkpoint evidence, fresh acknowledgments and a bounded ROOT release; existing guards and failures remain immutable.

Reproducible handoff

A separate private GitHub compact update is about 19 MB and includes code, reports, eight exact checkpoint files, configurations and derived TRAIN metadata. It is a declared subset; the raw 334 MB Cube TRAIN archive and other full-data assets are indexed externally. The original 5.868 GB full handoff and all previous PDFs/ZIPs remain unchanged. No installed runtime or transactional full-workspace snapshot is claimed.

This report is a fixed plan-328 snapshot for the morning checkpoint, not final closure. Copied source/JSON/CSV evidence and SHA manifests accompany it. Report generation made no model, sampler, environment or numerical payload queries. All final pages require rendered visual QA before delivery.